MULTIPLE NUISANCE BREAKS AND SPURIOUS PREDICTABILITY UNDER PERSISTENT REGRESSORS

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Abstract: We study inference on a stable predictive coefficient in a semiparametric predictive regression where the predictor may be highly persistent and the nonparametric nuisance component may undergo finitely many structural breaks at unknown dates. Imposing a stable nuisance function leaves a break-induced component in the orthogonalized empirical-likelihood score. When this component is aligned with the residualized persistent weight, a standard stable-nuisance procedure can report spurious evidence of predictability even when the predictive coefficient is zero. We propose a break-adaptive profile-sieve empirical likelihood. For each null value of the predictive coefficient, the procedure profiles a regime-specific sieve partition, residualizes both the outcome and the score weight against the resulting block sieve space, and forms a scalar empirical likelihood from the resulting orthogonalized score. The sieve formulation is used not because kernel methods are impossible, but because the block projection gives exact finite-sample cancellation of regime-specific nuisance components. Under finite-break, spacing, detectability, sieve-growth, and estimated-partition conditions, the empirical likelihood ratio evaluated at the estimated partition has a standard chi-square limit. The paper separates break-induced score drift from genuine predictive drift, reports Monte Carlo evidence on the persistence-break interaction, and gives an empirical commodity-weather application.

Key words and phrases: Empirical likelihood, nonparametric structural breaks, persistence, predictive regression, profile sieve, spurious predictability.

1. Introduction

Persistent predictors make predictive-regression inference fragile. In standard predictive regressions, least-squares limits vary with the persistence and tail behavior of the predictor. Weighted score empirical likelihood (EL) was introduced partly to avoid asking the researcher to classify the predictor's persistence before testing predictability; see, for example, Zhu, Cai and Peng (2014). This paper asks what happens when the same predictive regression also contains a flexible nonparametric control whose relation with the outcome is not stable over the sample.

The existing semiparametric profile-sieve construction treats the nuisance function as stable:

$$Y_t = \beta_0 X_{t-1} + m(W_{t-1}) + U_t.$$

This is a natural first step. It removes a nonlinear stationary control while preserving a weighted score for the persistent predictor. In long macroeconomic or financial samples, however, the response surface linking the control W_{t-1} to the outcome need not be stable. Weather exposures, policy regimes, measurement protocols, sector composition, or market microstructure can change the nuisance surface without changing the scientific question of interest: is the predictive coefficient β_0 stable and different from zero after allowing for such nuisance instability?

We therefore study

$$Y_t = \beta_0 X_{t-1} + \sum_{j=0}^{q_0} m_j(W_{t-1})\mathbf{1}\{k_{j,0} < t \leq k_{j+1,0}\} + U_t, \tag{1.1}$$

where $0 = k_{0,0} < k_{1,0} < \cdots < k_{q_0,0} < k_{q_0+1,0} = T$. The predictor X_t may be persistent, the coefficient β_0 is stable over the full sample, and all structural instability is placed in the nonparametric nuisance component. Thus the goal is not to estimate regime-specific predictive slopes and not to test whether β_t breaks. The goal is to test predictability after allowing the nuisance response to change at finitely many unknown dates.

The first contribution is to isolate the failure mechanism of a stable-nuisance procedure. If the true nuisance has breaks but the statistic projects only on a single global sieve space, the stable score has the exact decomposition

$$S_T^{\mathrm{st}}(\beta_0) = \frac{1}{\sqrt{T}} \boldsymbol{U}' \boldsymbol{M}_{K,0} \boldsymbol{w} + \frac{1}{\sqrt{T}} \boldsymbol{\mu}' \boldsymbol{M}_{K,0} \boldsymbol{w}$$

where $\mu_t = m_{j(t)}(W_{t-1})$ and $\boldsymbol{M}_{K,0}$ is the global stable-nuisance residual-maker. The second term is not a signal for β_0. It is the projection of an omitted low-frequency nuisance break onto the residualized persistent score direction. Persistent predictors do not mechanically create the signal, but they can amplify this alignment. Thus not every nuisance break causes false rejection; the relevant condition is that $T^{-1/2}\langle \boldsymbol{M}_{K,0}\boldsymbol{\mu}, \boldsymbol{M}_{K,0}\boldsymbol{w} \rangle$ is not negligible. This is a semiparametric analogue of the lesson in Perron (1989): an unmodeled low-frequency break can be mistaken for a stochastic or predictive feature of the series.

The second contribution is a break-adaptive score. For a candidate partition $\Lambda = (k_1, \ldots, k_q)$, define the block sieve matrix

$$\boldsymbol{Q}_{K,\Lambda} = \left[\boldsymbol{D}_0(\Lambda)\boldsymbol{P},\ \boldsymbol{D}_1(\Lambda)\boldsymbol{P},\ \ldots,\ \boldsymbol{D}_q(\Lambda)\boldsymbol{P} \right],$$

where $\boldsymbol{P}$ is the usual sieve matrix in W_{t-1}, and $\boldsymbol{D}_j(\Lambda)$ selects the j-th regime. Let

$$\boldsymbol{M}_{K,\Lambda} = \boldsymbol{I}_T - \boldsymbol{Q}_{K,\Lambda}(\boldsymbol{Q}'_{K,\Lambda}\boldsymbol{Q}_{K,\Lambda})^{\dagger}\boldsymbol{Q}'_{K,\Lambda}.$$

The proposed score is

$$\widehat{Z}_t(\beta, \Lambda) = \{\boldsymbol{M}_{K,\Lambda}(\boldsymbol{Y} - \beta \boldsymbol{X}_{lag})\}_t \{\boldsymbol{M}_{K,\Lambda}\boldsymbol{w}\}_t.$$

It preserves the same double-residualization geometry as the stable profile-sieve construction, but the nuisance space is enlarged from a single span of $P_K(W)$ to the regime-specific span generated by $\mathbf{1}(t \in I_j)P_K(W)$, $j = 0, \ldots, q$.

The third contribution is to make the partition part of the null-imposed inference procedure. For each candidate β, we estimate the partition by profiling the regime-specific sieve residual sum of squares. This avoids first estimating β and then treating the detected breaks as external preprocessing, which can contaminate the break criterion under persistent X_t. The confidence set is obtained by inversion of the break-adaptive EL ratio.

The fourth contribution is the asymptotic theory. The stable-nuisance method fails when the omitted-break component is non-negligible. The break-adaptive method has a known-partition Wilks theorem, a high-level partition-consistency condition, and an estimated-partition Wilks theorem. The proof uses the same finite-sample projection identity as the existing geometric manuscript, but with $\boldsymbol{P}$ and $\boldsymbol{M}_K$ replaced by $\boldsymbol{Q}_{K,\Lambda}$ and $\boldsymbol{M}_{K,\Lambda}$. The former geometric Wilks theorem becomes the known-partition core of the new paper, not the whole contribution.

The paper is related to four literatures. Weighted-score EL for predictive regressions provides persistence-robust inference for a stable lowdimensional predictive relation (Zhu, Cai and Peng, 2014); the scalar likelihood expansion builds on empirical likelihood theory (Owen, 1988, 2001; Qin and Lawless, 1994). Series approximations provide the sieve regularization used to remove nonlinear nuisance components (Newey, 1997). Multiple-break theory studies unknown structural changes in parametric regressions and related models (Andrews, 1993; Bai and Perron, 1998). Recent persistent predictive-regression break tests, such as Andersen and Varneskov (2021), target parameter instability and sup-Wald type diagnostics under persistence. Change-point algorithms such as narrowest-over-threshold and seeded binary segmentation (Baranowski, Chen and Fryzlewicz, 2019; Kovacs et al., 2023) provide efficient candidate generation. Our object is different: β_0 is stable, the breaks occur only in a nonparametric nuisance, and the inferential target is the post-nuisance-break EL statistic for β_0.

2. Model, Identification, and Break-Induced Predictivity

For a scalar array $\boldsymbol{a} = (a_1, \ldots, a_T)'$, write

$$\|\boldsymbol{a}\|_T^2 = T^{-1} \sum_{t=1}^{T} a_t^2, \qquad \|\boldsymbol{a}\|_\infty = \max_{t \leq T} |a_t|.$$

Unless stated otherwise, $O_p(\cdot)$ and $o_p(\cdot)$ for growing vectors and matrices are in Euclidean and spectral norm, respectively.

Let

$$\boldsymbol{Y} = (Y_1, \dots, Y_T)', \qquad \boldsymbol{X}_{lag} = (X_0, \dots, X_{T-1})',$$

and let $\boldsymbol{w} = (w_1, \dots, w_T)'$, where $w_t = w(X_{t-1})$ is a bounded saturated weight. The predictor obeys a persistence framework of the form

$$X_t = \theta + \rho_T X_{t-1} + \varepsilon_t, \tag{2.1}$$

covering stationary, local-to-unity, and unit-root cases through high-level oracle score conditions. The main model does not allow breaks in the predictor equation or in β_0. Those alternatives belong to parameter-instability testing; here they are deliberately excluded to keep the target as stable predictability after nuisance adjustment.

For a partition $\Lambda = (k_1, \dots, k_q)$, write $k_0 = 0$, $k_{q+1} = T$, and $I_j(\Lambda) = \{t : k_j < t \leq k_{j+1}\}$. The true partition is $\Lambda_0 = (k_{1,0}, \dots, k_{q_0,0})$. Define $j_\Lambda(t) = j$ when $t \in I_j(\Lambda)$. The true nuisance vector is

$$\mu_t = m_{j_{\Lambda_0}(t)}(W_{t-1}), \qquad \boldsymbol{\mu} = (\mu_1, \dots, \mu_T)'.$$

Assumption 1 (Finite nuisance breaks and smooth regimes)**.** *The number q_0 is fixed. For some $\epsilon > 0$,*

$$\min_{0 \leq j \leq q_0} (k_{j+1,0} - k_{j,0}) \geq \epsilon T.$$

Each m_j belongs to the same smoothness class $\mathcal{H}^s$. If $q_0 \geq 1$, the adjacent *breaks satisfy a detectability condition*

$$\Delta_T = \min_{0 \le j < q_0} \|m_{j+1} - m_j\|_{L^2(P_W)} > 0,$$

with $T\Delta_T^2$ large enough relative to the block sieve dimension and logarithmic search complexity. A sufficient fixed-break version is $T\Delta_T^2/(K \log T) \to \infty$. When $q_0 = 0$, this condition is void.

Assumption 2 (Identification after block residualization)**.** *For the true partition,*

$$T^{-1}\boldsymbol{w}'\boldsymbol{M}_{K,\Lambda_0}\boldsymbol{w} \xrightarrow{p} Q_w, \qquad P(Q_w > \underline{q}) = 1$$

for some $\underline{q} > 0$. Thus the residualized weighted predictive direction is not absorbed by the regime-specific nuisance space.

The stable procedure uses the global projection $\boldsymbol{M}_{K,0}$, where 0 denotes the no-break partition. Its null-imposed score numerator is

$$S_T^{\mathrm{st}}(\beta_0) = T^{-1/2}(\boldsymbol{Y} - \beta_0 \boldsymbol{X}_{lag})'\boldsymbol{M}_{K,0}\boldsymbol{w}.$$

Proposition 1 (Stable-score decomposition)**.** *Under the model* (1.1)*, the stable-nuisance score satisfies the exact identity*

$$S_T^{\mathrm{st}}(\beta_0) = A_T^{\mathrm{st}} + B_T,$$

where

$$A_T^{\mathrm{st}} = T^{-1/2}\boldsymbol{U}'\boldsymbol{M}_{K,0}\boldsymbol{w}, \qquad B_T = T^{-1/2}\boldsymbol{\mu}'\boldsymbol{M}_{K,0}\boldsymbol{w}.$$

The term B_T is the stable-projection nuisance component left by imposing the no-break nuisance projection. When $q_0 = 0$, this component is asymptotically negligible under the projection-bias condition in Assumption 3; when $q_0 > 0$, a nonnegligible component is interpreted as break-induced.

Proposition 1 identifies the object that can be misread as predictability. If $B_T = o_p(1)$, the stable score may remain valid. If B_T is non-negligible, the stable statistic is centered at the wrong score. The condition is directional: a nuisance break with little projection on the residualized persistent weight need not distort the test.

Definition 1 (Break-induced spurious predictivity)**.** The stable nuisance specification generates break-induced spurious predictivity at β_0 if

$$T^{-1/2}\langle \boldsymbol{M}_{K,0}\boldsymbol{\mu}, \boldsymbol{M}_{K,0}\boldsymbol{w}\rangle_{\mathbb{R}^T} = B_T \neq o_p(1).$$

The condition is directional. A nuisance break with little projection on the residualized persistent weight need not distort the test.

For the stable no-break empirical-likelihood statistic, write

$$V_T^{\mathrm{st}} = T^{-1}\sum_{t=1}^{T} \widehat{Z}_t(\beta_0, 0)^2,$$

where $\widehat{Z}_t(\beta_0, 0)$ is the score in (3.5) computed under the no-break partition.

Theorem 1 (Omitted-break failure)**.** *Suppose the stable empirical-likelihood statistic admits*

$$\ell_T^{\mathrm{st}}(\beta_0) = \frac{\{S_T^{\mathrm{st}}(\beta_0)\}^2}{V_T^{\mathrm{st}}} + o_p(1),$$

with V_T^{st} bounded away from zero in probability. Define

$$a_T = \frac{A_T^{\mathrm{st}}}{(V_T^{\mathrm{st}})^{1/2}}, \qquad b_T = \frac{B_T}{(V_T^{\mathrm{st}})^{1/2}}.$$

If $a_T = O_p(1)$ and $|b_T| \to_p \infty$, then, for every fixed chi-square critical value c_α,

$$P\{\ell_T^{\mathrm{st}}(\beta_0) > c_\alpha\} \to 1.$$

If instead

$$(a_T, b_T) \xrightarrow{\mathrm{st}} (Z, B),$$

where $Z \sim N(0,1)$ is independent of the limiting sigma-field and B is measurable with respect to that sigma-field, then

$$\ell_T^{\mathrm{st}}(\beta_0) \Rightarrow (Z + B)^2.$$

Conditionally on B, the limit is $\chi_1^2(B^2)$. Thus a constant $B = \delta$ gives $\chi_1^2(\delta^2)$, a random B gives a mixture of noncentral chi-squares, and $B = 0$ almost surely gives the central χ_1^2 limit.

Remark 1 (Scope of omitted-break divergence)**.** The divergence part of Theorem 1 is conditional on the displayed stable empirical-likelihood quadratic expansion. With a fixed omitted nuisance break, B_T can be of order $\sqrt{T}$, and the null-style scalar expansion need not follow automatically. Establishing generic fixed-break divergence therefore requires a separate expansion under the misspecified stable nuisance projection.

3. Regime-Specific Sieve Geometry

Let $\boldsymbol{P}$ be the $T \times K$ sieve matrix whose t-th row is

$$\boldsymbol{P}_K(W_{t-1})' = (p_1(W_{t-1}), \ldots, p_K(W_{t-1})).$$

For a partition Λ, let $\boldsymbol{D}_j(\Lambda)$ be the diagonal matrix with diagonal entry $\mathbf{1}\{t \in I_j(\Lambda)\}$. Define

$$\boldsymbol{Q}_{K,\Lambda} = [\boldsymbol{D}_0(\Lambda)\boldsymbol{P},\, \boldsymbol{D}_1(\Lambda)\boldsymbol{P},\, \ldots,\, \boldsymbol{D}_q(\Lambda)\boldsymbol{P}]\,, \tag{3.1}$$

and

$$\boldsymbol{M}_{K,\Lambda} = \boldsymbol{I}_T - \boldsymbol{Q}_{K,\Lambda}(\boldsymbol{Q}'_{K,\Lambda}\boldsymbol{Q}_{K,\Lambda})^{\dagger}\boldsymbol{Q}'_{K,\Lambda}. \tag{3.2}$$

Here † denotes the Moore–Penrose inverse. Thus the definition is valid for every candidate partition; when the block Gram matrix is nonsingular, it coincides with the ordinary inverse. The block nuisance space is

$$\mathcal{N}_{K,\Lambda} = \left\{ \sum_{j=0}^{q} \mathbf{1}(t \in I_j(\Lambda))\boldsymbol{P}_K(W_{t-1})'\boldsymbol{c}_j : \boldsymbol{c}_j \in \mathbb{R}^K \right\}.$$

The matrix $\boldsymbol{M}_{K,\Lambda}$ is the orthogonal residual-maker for this empirical space.

For each β and Λ, the profiled residual is

$$\widehat{\boldsymbol{u}}(\beta, \Lambda) = \boldsymbol{M}_{K,\Lambda}(\boldsymbol{Y} - \beta \boldsymbol{X}_{lag}), \tag{3.3}$$

and the regime-orthogonalized weight is

$$\boldsymbol{w}^c(\Lambda) = \boldsymbol{M}_{K,\Lambda}\boldsymbol{w}. \tag{3.4}$$

The scalar score is

$$\widehat{Z}_t(\beta, \Lambda) = \widehat{u}_t(\beta, \Lambda) w_t^c(\Lambda). \tag{3.5}$$

Lemma 1 (Exact block projection identities)**.** *For every partition* Λ*,*

$$\boldsymbol{Q}'_{K,\Lambda}\boldsymbol{M}_{K,\Lambda} = \boldsymbol{0}, \qquad \boldsymbol{M}'_{K,\Lambda} = \boldsymbol{M}_{K,\Lambda}, \qquad \boldsymbol{M}^2_{K,\Lambda} = \boldsymbol{M}_{K,\Lambda}.$$

Consequently, for every regime-specific sieve component $\boldsymbol{Q}_{K,\Lambda}\boldsymbol{c}$*,*

$$(\boldsymbol{Q}_{K,\Lambda}\boldsymbol{c})'\boldsymbol{M}_{K,\Lambda}\boldsymbol{w} = 0.$$

The empirical likelihood for (β, Λ) is

$$L_T(\beta, \Lambda) = \sup\left\{\prod_{t=1}^{T}(T\pi_t) : \pi_t \geq 0,\ \sum_{t=1}^{T}\pi_t = 1,\ \sum_{t=1}^{T}\pi_t \widehat{Z}_t(\beta, \Lambda) = 0\right\}, \tag{3.6}$$

$$\ell_T(\beta, \Lambda) = -2\log L_T(\beta, \Lambda) = 2\sum_{t=1}^{T}\log\{1 + \widehat{\lambda}(\beta, \Lambda)\widehat{Z}_t(\beta, \Lambda)\}, \tag{3.7}$$

where $\widehat{\lambda}(\beta, \Lambda)$ solves the scalar EL multiplier equation.

Remark 2 (Why sieve rather than kernel)**.** Kernel methods are possible in principle through undersmoothing, bias correction, cross-fitting, or Robinson-type residualization. The reason for using a sieve here is narrower: the EL proof needs a residual-maker that is an exact projection after regime-specific nuisance functions are introduced. A generic kernel smoother $\boldsymbol{S}_h$ need not satisfy $\boldsymbol{S}_h' = \boldsymbol{S}_h$ or $\boldsymbol{S}_h^2 = \boldsymbol{S}_h$, so $\boldsymbol{I} - \boldsymbol{S}_h$ is not generally an orthogonal residual-maker for the nuisance space. The sieve formulation reduces the proof burden to sieve approximation error and partition-estimation error, while retaining the finite-sample cancellation in Lemma 1.

4. Learning Multiple Nuisance Breaks

For an interval $[s, e]$, define the null-imposed segment cost

$$\mathcal{C}_K(s, e; \beta) = \min_{\boldsymbol{c} \in \mathbb{R}^K} \sum_{t=s}^{e} \{Y_t - \beta X_{t-1} - \boldsymbol{P}_K(W_{t-1})'\boldsymbol{c}\}^2. \tag{4.1}$$

For a partition $\Lambda = (k_1, \ldots, k_q)$,

$$RSS_K(\beta, \Lambda) = \sum_{j=0}^{q} \mathcal{C}_K(k_j + 1, k_{j+1}; \beta). \tag{4.2}$$

Let $\mathcal{L}_q(\epsilon)$ be the set of q-break partitions satisfying the minimum spacing condition. The theoretical profile estimator is

$$\widehat{\Lambda}_q(\beta) = \arg \min_{\Lambda \in \mathcal{L}_q(\epsilon)} RSS_K(\beta, \Lambda). \tag{4.3}$$

If the number of breaks is not fixed in advance, define

$$\widehat{q}(\beta) = \arg\min_{0 \le q \le \bar{q}} \left\{ RSS_K(\beta, \widehat{\Lambda}_q(\beta)) + \mathrm{pen}_T(q, K) \right\}, \tag{4.4}$$

and set

$$\widehat{\Lambda}(\beta) = \widehat{\Lambda}_{\widehat{q}(\beta)}(\beta).$$

The break-adaptive statistic is

$$\ell_T^{BA}(\beta) = \ell_T\{\beta, \widehat{\Lambda}(\beta)\}. \tag{4.5}$$

Confidence sets are obtained by inversion:

$$\mathcal{C}_{1-\alpha} = \{\beta : \ell_T^{BA}(\beta) \le \chi^2_{1,1-\alpha}\}.$$

The global minimization in (4.3) is the proof object. In computation, one can generate candidate break dates with seeded binary segmentation, narrowest-over-threshold, or related multiscale devices, using the split gain

$$G_K(s, e, b; \beta) = \mathcal{C}_K(s, e; \beta) - \mathcal{C}_K(s, b; \beta) - \mathcal{C}_K(b + 1, e; \beta).$$

The final selection is then based on the same profile-sieve RSS and information criterion. These algorithms are computational devices, not the statistical novelty.

5. Asymptotic Theory

Let $K_q = (q+1)K$ denote the block sieve dimension. All limits are along $K = K_T \to \infty$ with fixed q_0, unless stated otherwise.

Assumption 3 (Block-sieve rates and projection bias)**.** *For the true partition and for each regime j, there exists $\boldsymbol{c}_{j,K}$ such that*

$$m_j(W_{t-1}) = \boldsymbol{P}_K(W_{t-1})'\boldsymbol{c}_{j,K} + r_{j,K,t},$$

with $\max_j \|\boldsymbol{r}_{j,K}\|_T = O_p(a_K)$ *and* $\max_j \|\boldsymbol{r}_{j,K}\|_\infty = O_p(a_K)$*. Let $\boldsymbol{r}_0$ denote the stacked approximation-error vector under Λ_0. The directional projection bias satisfies*

$$T^{-1/2}\boldsymbol{r}_0'\boldsymbol{M}_{K,\Lambda_0}\boldsymbol{w} = o_p(1).$$

The block Gram matrix $T^{-1}\boldsymbol{Q}'_{K,\Lambda_0}\boldsymbol{Q}_{K,\Lambda_0}$ has eigenvalues bounded away from zero and infinity with probability tending to one. If $\zeta_K = \max_t \|\boldsymbol{P}_K(W_{t-1})\|$*, then*

$$a_K(1+\zeta_K) \to 0, \qquad a_K(1+\zeta_K)^2 = o(\sqrt{T}),$$

and

$$\frac{(q_0+1)K}{\sqrt{T}} \to 0, \qquad \zeta_K\sqrt{\frac{(q_0+1)K}{T}} \to 0.$$

Remark 3 (Projection bias versus undersmoothing)**.** The condition $T^{-1/2}\boldsymbol{r}_0'\boldsymbol{M}_{K,\Lambda_0}\boldsymbol{w} = o_p(1)$ only requires the nuisance approximation error to be negligible in the residualized score direction. It is weaker than the full root-T undersmoothing condition $\sqrt{T}a_K \to 0$. A sufficient primitive route is a product-rate condition $\sqrt{T}\, a_{m,K} a_{w,K} \to 0$, where $a_{m,K}$ approximates the nuisance functions and $a_{w,K}$ approximates the conditional mean of the weight w_t given W_{t-1}.

Assumption 4 (Known-partition oracle score)**.** *With* $\boldsymbol{v}_0 = \boldsymbol{M}_{K,\Lambda_0}\boldsymbol{w}$, *define* $\widetilde{\boldsymbol{U}}_0 = \boldsymbol{M}_{K,\Lambda_0}\boldsymbol{U}$ *and*

$$A_T^{(q)} = T^{-1/2}\sum_{t=1}^{T} U_t v_{t,0}, \qquad V_T^{(q)} = T^{-1}\sum_{t=1}^{T} U_t^2 v_{t,0}^2.$$

There exist a random scale η_q*, a constant* $c_\eta > 0$*, and a standard normal* Z*, independent of the limiting sigma-field generating* η_q*, such that*

$$A_T^{(q)} \xrightarrow{\text{st}} \eta_q Z, \qquad V_T^{(q)} \xrightarrow{p} \eta_q^2, \qquad P(\eta_q \geq \sqrt{c_\eta}) = 1.$$

In addition,

$$\max_{t\leq T} |U_t v_{t,0}| = o_p(\sqrt{T}),$$

and the projection of the noise onto the block sieve is asymptotically negligible in the weighted score array:

$$T^{-1}\sum_{t=1}^{T}\{(\widetilde{U}_{0,t}v_{t,0})^2 - (U_t v_{t,0})^2\} = o_p(1), \qquad \max_{t\leq T} |(\widetilde{U}_{0,t} - U_t)v_{t,0}| = o_p(\sqrt{T}).$$

Finally, the feasible true-partition score array satisfies the scalar empirical-

likelihood convex-hull condition

$$P\left\{\min_{t\le T}\widehat{Z}_t(\beta_0,\Lambda_0) < 0 < \max_{t\le T}\widehat{Z}_t(\beta_0,\Lambda_0)\right\}\to 1.$$

This condition is imposed on the feasible array $\widehat{Z}_t(\beta_0,\Lambda_0)$, not only on the oracle array $U_t v_{t,0}$.

Theorem 2 (Known-partition Wilks theorem)**.** *Under Assumptions 1, 2, 3, and 4, under $H_0:\beta=\beta_0$,*

$$\ell_T(\beta_0,\Lambda_0)\Rightarrow\chi^2_1.$$

Theorem 2 is the block-sieve version of the original geometric Wilks result. The key change is that the oracle weight is no longer a globally centered weight. It is $v_{t,0}=(\boldsymbol{M}_{K,\Lambda_0}\boldsymbol{w})_t$, the residualized weight after projecting on the regime-specific nuisance space.

For the fixed-order analysis, define

$$\widetilde{\Lambda}=\widehat{\Lambda}_{q_0}(\beta_0),\qquad \widetilde{\Lambda}=(\widetilde{k}_1,\ldots,\widetilde{k}_{q_0}).$$

The unknown-order estimator $\widehat{\Lambda}(\beta_0)$ is used only in Corollary 1.

Assumption 5 (Fixed-order partition localization)**.** *For the true number of breaks,*

$$\max_{1\le j\le q_0}|\widetilde{k}_j-k_{j,0}|=O_p(r_T),\qquad r_T=o(\sqrt{T}).$$

Equivalently, the number of fixed-order observations assigned to the wrong regime satisfies

$$|\mathcal{M}_T| = \left|\{t : j_{\widetilde{\Lambda}}(t) \neq j_{\Lambda_0}(t)\}\right| = o_p(\sqrt{T}).$$

This localization condition does not by itself imply score equivalence, because moving boundary observations also perturbs the fitted regime coefficients outside $\mathcal{M}_T$.

Assumption 6 (High-level fixed-order estimated-score equivalence)**.** *Under $H_0 : \beta = \beta_0$,*

$$T^{-1/2} \sum_{t=1}^{T} \{\widehat{Z}_t(\beta_0, \widetilde{\Lambda}) - \widehat{Z}_t(\beta_0, \Lambda_0)\} = o_p(1), \tag{5.1}$$

$$T^{-1} \sum_{t=1}^{T} \{\widehat{Z}_t(\beta_0, \widetilde{\Lambda})^2 - \widehat{Z}_t(\beta_0, \Lambda_0)^2\} = o_p(1), \tag{5.2}$$

$$\max_{t \le T} |\widehat{Z}_t(\beta_0, \widetilde{\Lambda})| = o_p(\sqrt{T}). \tag{5.3}$$

Zero lies in the interior of the scalar empirical-likelihood convex hull for the fixed-order estimated-partition score array with probability tending to one. This is a high-level score-equivalence condition; Assumption 5 alone is not claimed to prove it.

Theorem 3 (Fixed-order estimated-partition Wilks theorem)**.** *Under the assumptions of Theorem 2 and Assumption 6,*

$$\ell_T(\beta_0, \widetilde{\Lambda}) \Rightarrow \chi^2_1.$$

This theorem is high-level: it proves that fixed-order estimated-score equivalence implies empirical-likelihood equivalence, not that the profile-sieve partition estimator satisfies Assumption 6 from primitive conditions.

Corollary 1 (Unknown number of breaks)**.** *If $P\{\widehat{q}(\beta_0) = q_0\} \to 1$ and Assumption 6 holds on the fixed-order event, then*

$$\ell_T^{BA}(\beta_0) \Rightarrow \chi_1^2.$$

Theorem 4 (Local predictive alternatives)**.** *Let $\beta_T = \beta_0 + d_T$, where $d_T \to 0$. Define*

$$g_{t,T} = d_T(\boldsymbol{M}_{K,\Lambda_0}\boldsymbol{X}_{lag})_t v_{t,0}.$$

Suppose

$$T^{-1/2}\sum_{t=1}^{T} g_{t,T} \xrightarrow{p} \eta_q \delta_h, \tag{5.4}$$

$$T^{-1}\sum_{t=1}^{T} g_{t,T}^2 = o_p(1), \tag{5.5}$$

$$\max_{t \le T} |g_{t,T}| = o_p(\sqrt{T}), \tag{5.6}$$

with δ_h fixed, and set $Z_{t,T}^{\text{loc}} = U_t v_{t,0} + g_{t,T}$. Under the local data-generating *law β_T, assume the feasible true-partition score satisfies*

$$T^{-1/2}\sum_{t=1}^{T}\{\widehat{Z}_t(\beta_0,\Lambda_0)-Z_{t,T}^{\mathrm{loc}}\}=o_p(1), \tag{5.7}$$

$$T^{-1}\sum_{t=1}^{T}\{\widehat{Z}_t(\beta_0,\Lambda_0)^2-(Z_{t,T}^{\mathrm{loc}})^2\}=o_p(1), \tag{5.8}$$

$$\max_{t\le T}|\widehat{Z}_t(\beta_0,\Lambda_0)-Z_{t,T}^{\mathrm{loc}}|=o_p(\sqrt{T}). \tag{5.9}$$

Also assume that, under β_T, the feasible true-partition score array satisfies

$$P_{\beta_T}\left\{\min_{t\le T}\widehat{Z}_t(\beta_0,\Lambda_0)<0<\max_{t\le T}\widehat{Z}_t(\beta_0,\Lambda_0)\right\}\to 1.$$

Assume, in addition, that the fixed-order estimated-score equivalence in Assumption 6 holds under β_T, with all o_p statements evaluated under the local law. Then, when the statistic is evaluated at β_0,

$$\ell_T(\beta_0,\widetilde{\Lambda})\Rightarrow\chi_1^2(\delta_h^2).$$

If $P\{\widehat{q}(\beta_0)=q_0\}\to 1$, the same limit holds for $\ell_T^{BA}(\beta_0)$. The noncentrality parameter is generated by the predictive drift d_T. It does not contain the nuisance-break drift B_T, which is removed by the regime-specific projection.

Proof-dependency map

The theory deliberately separates the geometry proved in the paper from high-level persistence and segmentation inputs.

Step	Direct paper or supplement argument	High-level input
Stable-method failure	Score decomposition and scalar EL expansion around the contaminated score.	Stable oracle expansion and non-negligible B_T.
Known-partition Wilks	Block projection identities, projection-bias replacement, and scalar EL expansion.	Persistent oracle mixed-normal score for $U_t v_{t,0}$, feasible convex hull, and negligible block projection of noise.
Partition learning	Profile-sieve RSS and candidate-segmentation construction.	Spacing, detectability, and $r_T = o(\sqrt{T})$ localization.
Estimated-partition Wilks	Likelihood transfer from the fixed-order estimated score to the true-partition score.	High-level estimated-score equivalence; localization alone is not asserted to prove it, plus consistent $\widehat{q}$ for unknown order.

6. Monte Carlo Evidence

The Monte Carlo experiment is designed to isolate the mechanism in Theorem 1. The core DGP is

$$Y_t = \beta X_{t-1} + m_{j(t)}(W_{t-1}) + U_t, \qquad X_t = \rho X_{t-1} + \varepsilon_t,$$

with W_t stationary and correlated with the innovation driving X_t. The nuisance function is stable when $q_0 = 0$. In the break design, $q_0 = 1$, the break is at mid-sample, and the second regime adds a level shift of 1.25 to the same smooth baseline function. The predictive null is $\beta = 0$, so rejection by the stable statistic reflects the omitted-break score component rather than genuine predictability.

The reported procedures are the stable sieve EL, the oracle-break sieve EL, and the estimated-break sieve EL. The estimated procedure selects between zero and one nuisance break by the null-imposed profile-sieve information criterion. The persistence grid is $\rho \in \{0.50, 0.90, 0.98, 1.00\}$, with $T = 240$, $K = 4$, and 300 replications per design point.

Figure 1 and Table 1 report rejection rates at the five percent nominal level. Under $q_0 = 0$, all three procedures remain close to nominal size, apart from the familiar finite-sample strain at the unit-root endpoint. Under $q_0 = 1$, the stable statistic increasingly rejects as persistence rises: the

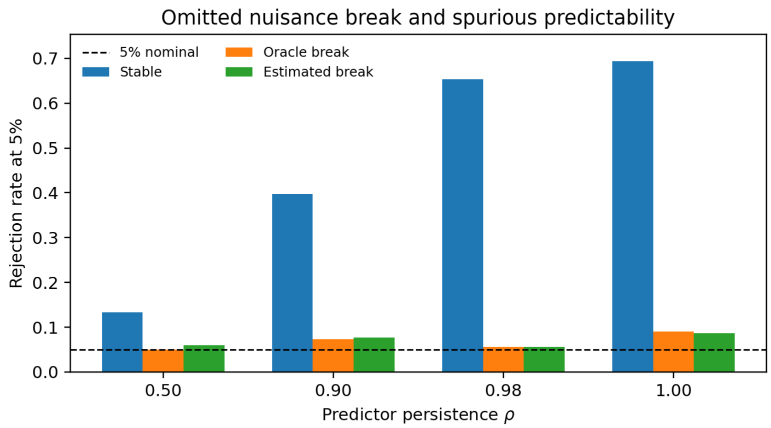

Figure 1: Monte Carlo rejection rates when the nuisance function has one omitted level break. The stable statistic confounds the omitted nuisance break with predictability as persistence increases; the oracle and estimated-break statistics remove the break-induced score component.

rejection rate rises from 0.133 at $\rho = 0.50$ to 0.693 at $\rho = 1.00$. The oracle and estimated-break procedures remain near the nominal level throughout the same design. The estimated partition selects one break in every break-design replication, with mean absolute break-date error between 1.2 and 2.0 observations.

7. Empirical Application

The empirical application uses daily cocoa returns and Ghana weather anomalies. The sample contains 6,669 daily observations from October

Table 1: Monte Carlo rejection rates under nuisance breaks

q_0	ρ	Stable	Oracle	Estimated	$P(\widehat{q}=1)$	Break error
0	0.50	0.047	0.047	0.043	0.033	–
0	0.90	0.037	0.037	0.040	0.037	–
0	0.98	0.060	0.060	0.057	0.017	–
0	1.00	0.090	0.090	0.093	0.033	–
1	0.50	0.133	0.050	0.060	1.000	1.5
1	0.90	0.397	0.073	0.077	1.000	2.0
1	0.98	0.653	0.057	0.057	1.000	1.2
1	1.00	0.693	0.090	0.087	1.000	1.4

Notes: $T = 240$, $K = 4$, 300 replications in the submission run unless the script is called with a different replication count. The break design has a level shift of 1.25 at mid-sample. The estimated procedure selects between zero and one nuisance break by the null-imposed profile-sieve information criterion.

4, 1994 to November 28, 2024. Let R_t denote the ICCO cocoa log return in percentage points, let X_{t-1} be the standardized lagged log cocoa price, and let the weather index be the first principal component of standardized Ghana precipitation and temperature anomalies. The fitted model is

$$R_t = \beta X_{t-1} + m_{j(t)}(\text{Weather}_{t-1}) + U_t.$$

The same null-imposed profile-sieve information criterion used in the simulation is applied with $q \leq 3$, a two-year minimum regime length, and a six-month candidate grid.

Table 2 reports the stable and break-adaptive EL statistics at $\beta = 0$. The information criterion selects $\widehat{q} = 0$, so the stable and break-adaptive statistics coincide: the EL ratio is 0.069 and the p-value is 0.793. Thus this application does not provide evidence of commodity-return predictability, nor does the conservative profile-sieve criterion find nuisance breaks large enough to alter the predictive conclusion. This is still an informative application because the procedure reports the absence of break-induced predictivity instead of forcing a break-adjusted story.

8. Conclusion

The paper studies a specific source of spurious predictability: multiple unknown breaks in a nonparametric nuisance component. A stable semiparametric procedure leaves a break-induced term in the score when the omitted nuisance break is aligned with the residualized persistent weight. The proposed break-adaptive sieve EL removes that component by profiling a regime-specific nuisance partition under each null value and residualizing both the outcome and the score weight against the resulting block

Table 2: Commodity-return predictability with Ghana weather controls

	Stable nuisance	Break-adaptive nuisance
EL ratio at $\beta = 0$	0.069	0.069
p-value	0.793	0.793
Score	-0.576	-0.576
Pred. strength	0.285	0.285
$\widehat{q}$	0	0

Notes: The sample has 6669 daily observations from 1994-10-04 to 2024-11-28. The weather index is the first principal component of precipitation and temperature anomalies. Estimated nuisance break dates: None.

sieve space. The previous profile-sieve geometry is retained as the known-partition core, while the new results add the stable-method failure theorem, partition learning, estimated-partition Wilks inference, and the empirical distinction between break-induced score drift and genuine predictive drift.

9. Supplementary Material

The online Supplement gives the omitted-break expansion, the projection-bias Wilks proofs, the high-level estimated-partition transfer, the shifted-array local-power argument, and implementation details.

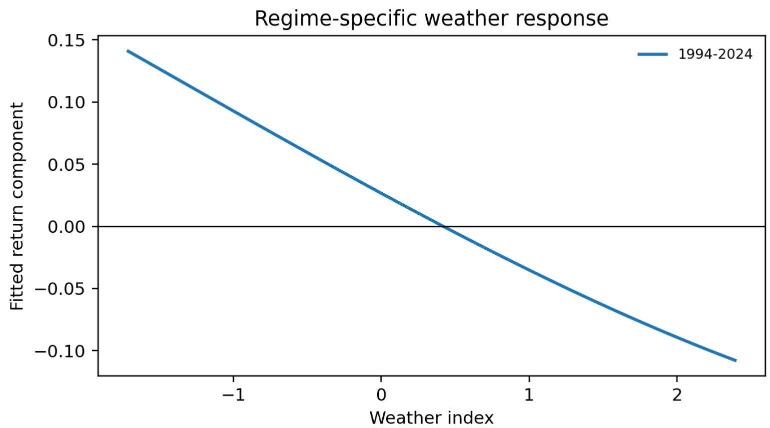

Figure 2: Estimated weather response for the Ghana cocoa application under the selected nuisance partition. The information criterion selects no nuisance break, so the displayed curve is the stable weather-index response.

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